Shakti Kumar

Data Analysis & Machine Learning · PhD Scholar, IIT Kanpur

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Profile

PhD scholar at IIT Kanpur (M.Tech, IIT Kanpur; B.Tech, Mechanical Engineering) with GATE 2024 Computer Science AIR 22,651. Cleans, analyses and models data in Python, SQL, Excel and Tableau, backed by IIT Kanpur machine-learning coursework (CS771A, MSE643), and explains results in plain language.

Skills

- **Data analysis:** Python (pandas, NumPy, matplotlib, seaborn), SQL and relational databases, Excel, Tableau, applied statistics, exploratory data analysis
- **Machine learning:** scikit-learn (linear models, decision trees, random forests, gradient boosting), XGBoost, cross-validation, GridSearchCV, feature engineering, SHAP
- **Tools:** Jupyter and Google Colab, Git and GitHub, MATLAB, LaTeX · **Languages:** English (professional), Hindi (native)

Projects

Airbnb NYC Bookings Analysis — github.com/Shaktikr/Capstone-EDA-project-Airbnb-bookings-analysis · individual capstone, 2023

- Exploratory analysis of 48,895 New York listings: missing values, IQR outlier removal on price, 18 visualisations and business conclusions. Manhattan averages $146 a night against $77 in the Bronx; Manhattan and Brooklyn hold 85% of listings.

ML Stress Prediction for Gyroid Lattices — github.com/Shaktikr/gyroid-fea-ml · MSE643 course project, IIT Kanpur

- Compared five regressors (linear, decision tree, random forest, gradient boosting, XGBoost) on 2,525 Abaqus simulation records with 5-fold cross-validation; a feature-engineered linear model won (test R^2 0.9999, MAE 0.14 MPa), confirmed with a bias–variance curve and GridSearchCV.

Machine Learning for Materials Data — github.com/Shaktikr/mse-computational-projects · 2026, built with AI coding assistance and verified

- Gradient-boosting model of creep-rupture life with SHAP explanations; phase classifier on a real 1,545-entry high-entropy-alloy dataset reaching 80% accuracy against 58% for empirical rules.

Sports-Car Price Data — github.com/Shaktikr/project-predicting-sports-car-prices · 2023

- Cleaned a messy price dataset (text-to-number conversion, missing values, duplicates), segmented makes and engine sizes, and compared price distributions.

Experience

AI Training Task Author (freelance) — Alignerr (Labelbox) · 2026 – present

- Authoring an engineering RL task for frontier AI models, with a reference solution and a deterministic Python grader tested through Git/CI.

Graduate Teaching Assistant — IIT Kanpur · Jul 2023 – May 2024; Dec 2025 – present

- Maintains attendance and marks records for a weekly laboratory batch in Excel; tutorials and rubric-based evaluation.

Teaching Faculty — GeeksforGurukul, Indore · Nov 2024 – May 2025

Doubt-Solving Educator — Unacademy · Aug 2021 – Jul 2022 · 200+ GATE video solutions

Education

- **PhD, Materials Science & Engineering (ongoing) — IIT Kanpur** · 2025 – present (AI & ML in Materials Engineering)
- **M.Tech, Aerospace Engineering — IIT Kanpur** · 2022 – 2024 (CS771A Machine Learning)
- **B.Tech, Mechanical Engineering — REC Banda** · 2018 – 2022 · CPI 7.85
- **GATE 2024 (Computer Science & IT):** AIR 22,651 of 1,23,967 · **GATE 2021 (Mechanical):** AIR 8,664 of 1,20,594 (top 7.2%)

Certifications

- **AlmaBetter, 2023:** Applied Statistics; Business Analysis; Data Analysis Using Excel; SQL (Getting Started, Data Preparation, In-Built Functions); Relational Databases; Tableau
- **LinkedIn Learning, 2023:** Become a Data Scientist (23 h); Career Essentials in Data Analysis (Microsoft & LinkedIn); Getting Started with Python; Learning Data Analytics; AI Foundations: ML